

① What is ML & how it is different from traditional field.

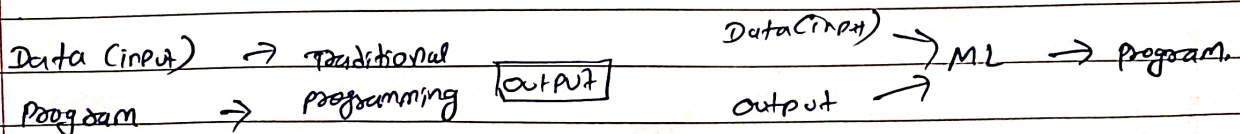
→ ML stands for Machine Learning which is a subset of artificial intelligence (AI) that focuses on creating systems that can learn and improve from data and make decisions without being explicitly programmed for every specific task.

Three main types

- i) Supervised learning: The system learns from labeled data with input and output.
- ii) Unsupervised learning: System finds hidden patterns in data without being given specific labels.
- iii) Reinforcement learning: System learns through trial and error, receiving rewards or penalties for its actions.

ML VS Traditional approach

- Approach: ML learns from ex vs traditional programming use rules.
- Problem-solving: ML handles complex, hard to define problem vs traditional method need clear step-by-step logic.
- Data needs: ML requires lots of training vs we do exactly what they rely on human.
- Performance: ML improves over time with more data & traditional system stays consistent.
- Adaptability: ML adapt new situation, traditional system do exactly what they are coded for.



② What are the components we must know for Learning ML.

→ Some basic components of ML we must know are listed below:

i) Mathematics

- It includes rules and regulation and basic knowledge in probability, graphs, statistics, calculus, matrix, linear algebra etc.

ii) Data

→ Huge data set on which we are able to train an algorithm and models.

iii) Model

→ The trained system that makes predictions after learning from data.

iv) Deployment & Evaluation

→ Implementing the trained model in real-world applications to make actual predictions and measure model performance.

v) Algorithms

vi) Feedback loop

③ Difference between AI & ML?

→ Artificial Intelligence	Machine Learning
i) Replicates human like behaviour	subset of AI focused on learning from data
ii) Rule-based, expert systems, robotics, ML.	Data-driven, statistical learning algorithms.
iii) General intelligence problems	Pattern recognition and predict tasks.
iv) Started in 1950s	Emerged later as AI subset.
v) can be rule / learning based	Always learning-based.
vi) Example: chess programs, chatbots, robotics, expert systems	Example: Recommendation engines, images recognition, spam detection.

④ Some examples of ML?

→ Youtube recommendation system

- Face recognition
- Image classification
- Spam email filter
- Healthcare diagnosis
- Fraud detection
- Self-driving cars
- E-commerce

Advantages of ML

- i) Easily identifies trends and patterns
- ii) No human intervention needed
- iii) Continuous Improvement
- iv) Handling multi-dimensional and multi-variety data.
- v) wide Applications.

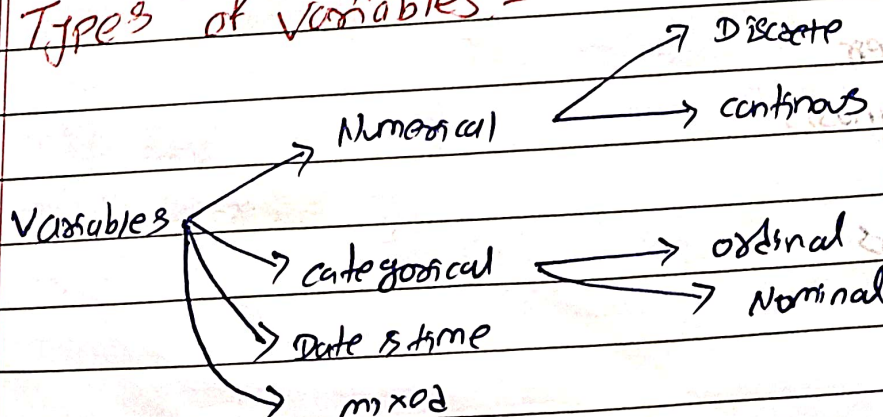
Disadvantage of ML

- i) Data Acquisition
- ii) Time and Resource
- iii) Interpretation of Results
- iv) High error susceptibility.

Data:

- Data is raw fact or figure or entity when activities in the organization take place, the effect of these activities need to be recorded which is known as data.

Types of variables:-



Numerical variable (int, float)

↓ Discrete ↓ Continuous

- perfect number - decimal numbers
- Repeat - multiple data

eg: 1, 2, 3, 1, 2, 3

eg: 1.1, 1.2, 1.3, 1.5, 1.7

categorical variables (string)

↓ Ordinal

↓ Nominal

- Follows order - No interconnection
- Sequence pattern - Random data

eg: size (S, L, XL, XXL)

eg: Python, Java, Language

Date & time

→ Date & time can contain date only, time only.

Mixed variables

→ variables which contains numbers and categories data.

Managing & storing data

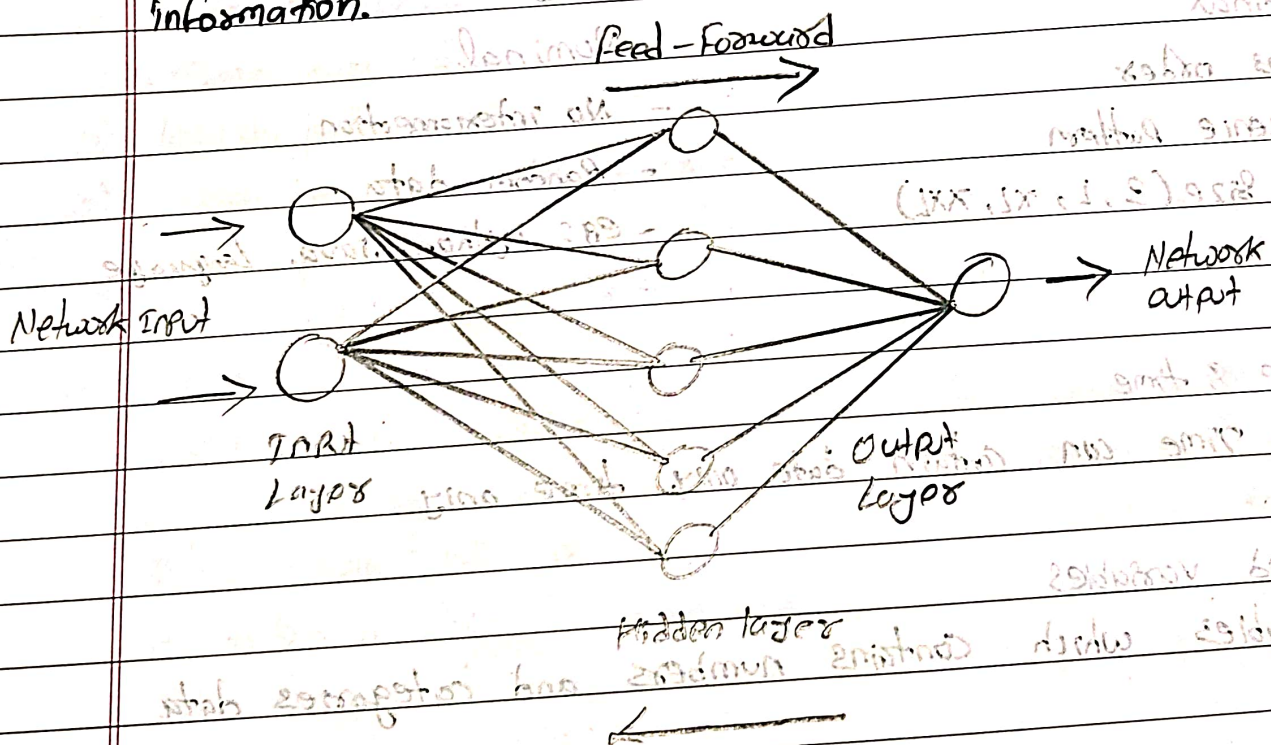
- i) collect relevant data.
- ii) pre process and clean data
- iii) Annotate data
- iv) split data set
- v) Analyze the data set
- vi) Document the dataset
- vii) store and manage the data.

Network:-

A network is a collection of connected nodes organised in layers that learns patterns from data to make predictions or decisions.

Neural Network:-

- A Neural Network is a ML model that's inspired by how human brain work.
- It's made up of nodes that process and pass information.



Input Layer:-

- Takes in the data (like image, numbers, text)
- Each node = one feature of the input

Hidden Layer (can be 1 or more)

- where real learning happens.
- Each neuron, does some math (activation function, weights, biases) and passes the result of the next layer.
- more layers = deeper network = Deep learning.

Output layer

- Gives you the prediction (like cat or dog, spam or not spam, etc).

Formula:

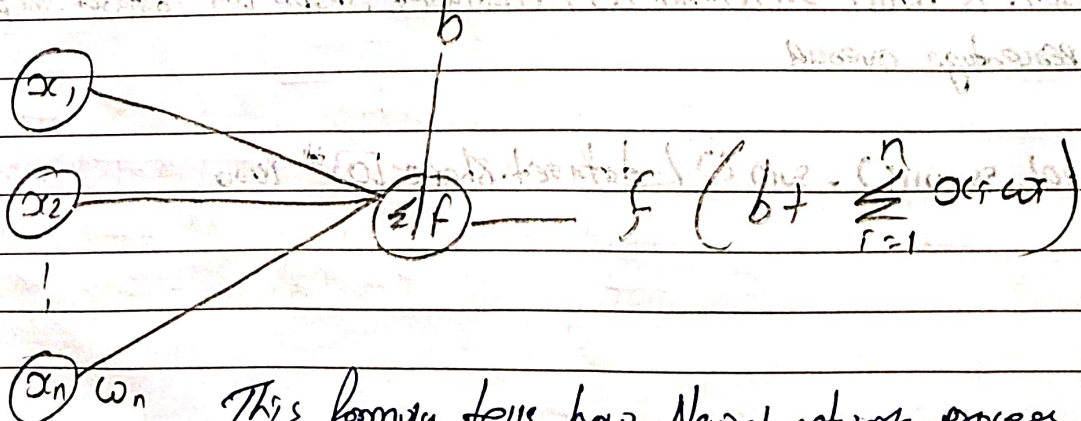
$$z = \sum_{i=1}^n w_i x_i + b$$

where,

- z is the weighted sum of inputs of the neuron
- w_i is the weight associated with the i -th input
- x_i is the i -th input to the neuron.
- b is the bias term.
- n is the total no of inputs

The output is then calculated of the neuron by applying the activation function (σ) to this weighted sum:

$$a = \sigma(z)$$



This formula tells how Neural network process information.

Data cleaning:

→ Data cleaning is the process of preparing the data for Analysis / AI / ML / DL by removing or modifying data that is incorrect, incomplete, irrelevant, duplicated, or improperly formatted.

Most important Steps:

- Handling missing data
- outlier detection and handling
- Data Scaling and Transformation
- Encoding categorical variables
- Handling Duplicates
- Dealing with Inconsistent Data

import pandas as pd

dataset = pd.read_csv('path')

dataset.head(5)

dataset.shape

dataset.isnull().sum() # table output with null

dataset.isnull().sum().sum # total no of null

$(dataset.isnull().sum().sum() / (dataset.shape[0] * dataset.shape[1])) * 100$

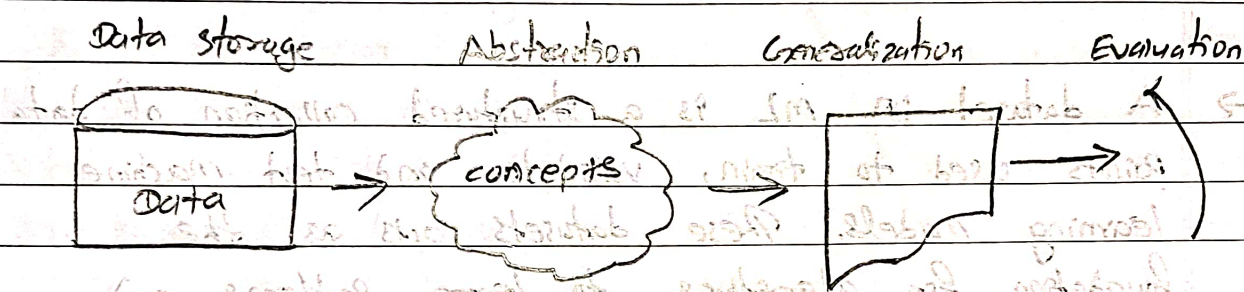
percentage overall

$dataset.isnull().sum() / dataset.shape[0] * 1000$

Components of Machine learning?

- The learning process whether by a human or a machine, can be divided into four components, namely, data storage, abstraction, generalization and evaluation.

Figure illustrates the various components and the steps involved in the learning process.



1) Data storage

Facilities for storing and retrieving huge amount of data are an important component of the learning process. Humans and computers like utilize data storage as a foundation for advanced reasoning.

2) Abstraction

- The second component of the learning process is known as abstraction. Abstraction is the process of extracting knowledge about stored data.

3) Generalization:

- The third component of the learning process is known as generalization. The term generalization describes the process of turning the knowledge about stored data into form that can be utilized for future action.

4) Evaluation:

- Evaluation is the last component of the learning process. It is the process of giving feedback to the users to measure the utility of the learned knowledge.

Data Set and Machine Learning:

- A dataset in ML is a structured collection of data points used to train, validate, and test machine learning models. These datasets act as the foundation for algorithms to learn patterns and make informed predictions or decisions.

Handling datasets:

- A comprehensive process. Handling datasets for machine learning involves several critical steps to ensure data quality and suitability for modeling.

Define Objective,

Collect Relevant data,

Preprocess and clean

Annotate data,

Split the Dataset,

Analyze the Dataset,

Document the Dataset

Store and manage data.

Feature sets: enhancing model performance

- A feature set is collection of traits, characteristics, or attributes used to represent an object or phenomenon numerically for M.L algorithms. Features are the input variables that the model uses to make predictions.

Importance of feature sets

- Data Representation
- Feature Engineering
- Dimensionality Reduction
- Interpretability

Data Set Division?

- To ensure the reliability and generalization of a M.L model, the dataset is typically split into three subsets.

Training set: The largest portion of the dataset, used to teach the model and fit its parameters by identifying patterns and relationships within the data.

Validation set:

- used to evaluate and fine-tune the model's performance during training, helping adjust hyperparameters and prevent overfitting.

Test set:

- provides an unbiased evaluation of the final trained model on unseen data, approximating its real-world performance.

Cross-validation:

→ It is a technique used to assess a model's predictive performance and ensure it generalizes well to unseen data, particularly crucial with limited datasets.

K folds cross-validation:

- 1) Partition the data: Divide the dataset into K subsets (folds) of approximately equal size.
- 2) Train and Test: In each iteration, train the model on $K-1$ folds and test on the remaining fold.
- 3) Calculate performance metrics: Assess the model's performance on the test set for each iteration.
- 4) Aggregate Results: Average the performance metrics from all iterations to get an overall assessment of the model's generalization capability.

"Cross-validation helps identify and mitigate issues like overfitting and selection bias, leading to more reliable and accurate models, especially valuable when data is scarce."

Networks Evaluation:

- Evaluating machine learning model is crucial for determining their effectiveness and reliability. This involves using evaluation metrics, which are quantitative measure that assess the model's performance.

The choice of metrics depends on the type of problem, such as classification or regression. Some common evaluation metrics include:

* Accuracy:

- The proportion of correct predictions out of the total predictions.

* Precision:

The proportion of positive predictions that were actually correct.

* Recall (sensitivity):

The proportion of actual positive cases that were correctly identified

* Confusion matrix:

- A table that summarizes the predictions of a classification model, showing true positives, true negatives, false positives, and false negatives.

* Roc AUC Curve (Receiver Operating Characteristic - Area under the curve):

- A curve that illustrates the performance of a binary classifier system as its discrimination threshold is varied.

* RMSE (Root mean squared error):

- Commonly used for regression problems to measure the magnitude of the errors.

2. Basics of Machine Learning

Machine Learning is a subset of Artificial Intelligence that allows computers to learn from data and make decisions without being explicitly programmed.

Essentially, ML algorithms identify patterns in data and use those patterns to predict or classify new, similar data. This ability to learn from data and improve over time is what makes ML so powerful.

Applications of machine learning

Machine Learning is seamlessly integrated into our daily lives and used across diverse industries.

1) Traffic Management

- Machine learning can be used to predict and control traffic flow in real time.

Example:

In emergency situations such as transporting a donor heart for a transplant from one state to another, the donor hospital can coordinate with the traffic authorities to create a "green corridor" - a route where traffic is completely cleared to ensure fast, uninterrupted travel.

ML systems can analyze live traffic data, predict congestion points and suggest the best possible route while keeping traffic signals synchronized, ensuring the heart reaches the patient's hospital in time.

2) Health care:

- Machine learning plays a crucial role in assisting doctors during medical procedures. AI-powered assistants can monitor an operation in real time, anticipate which tools the doctor will need next, and provide instant access to relevant medical data. These systems can also analyze patient reports and scans with high accuracy, helping doctors identify health problems earlier and more precisely.
- With the integration of robotics and telemedicine, doctors can even perform surgeries on patients in another country, using real-time data transfer and AI-driven guidance.

3) Weather Forecasting

- Machine learning models are widely used to predict upcoming weather conditions, such as whether it will rain in the next few days. These systems analyze vast amounts of data, including satellite images, ocean patterns, wind speed, and atmospheric pressure.

Example:

Changes in water pressure and temperature in the Bay of Bengal can signal the formation of cyclones or heavy rainfall. ML algorithms process this data, optimize prediction accuracy, and provide reliable forecasts, which are then used by weather apps to update users in real time.

IV) Speech Recognition: (Image recognition)

- Biometric data and retina scans of the students and after that they don't have to mark attendance. The cameras in various campus automatically detects faces and does all the things.

V) Marketing:

- Government can create new policy according to the available data collected from all the sub-authorities about the production of ex tomatoes and we can analyze the production.

VI) Economic Growth Banking:

- Analysis of the market and predicting the future progress and outcomes. Money Distribution system by the centralized authority with the required data from the respective Banks.

VII) Agriculture:

- Analysis the production state and the proper selection of seeds (e.g. seed no) so that the cultivation can be boosted considering all conditions like land, weather and production rate and season with crop rotations.

VIII) Sports:

- Like in cricket capture the Blower's activities and according to Blower we train the Bat's man.